THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

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KWIMBA DISTRICT COUNCIL

FORM FOUR MOCK EXAMINATIONS – 2023

CHEMISTRY

SUBJECT CODE: 032 **Time: 3 Hours**

INSTRUCTIONS:

- This paper consists of section A, Band C with a total of 14 questions.
- Answer all questions in section A and B and one (1) questions from section C.
- Unauthorized materials are not allowed in the examination room
- The following constants may be used

Atomic masses: $H=1, Cl=35.5, C=12, O=16 N=14, Cu=64, Ag=108$
$Ca=40, S=32, Mn=55$

- Avagadro's number 6.02×10^{23}
- 1 Farady=96,500 Coulombs
- G.M.V at s.t.p $22.4 dm^3$
- 1 litre=$1 dm^3$=$1000 cm^3$

SECTION A

1. For each of the following items (i-x) choose the correct answer given alternatives and write its letter in the answer booklets provided.

(i) Mwl. Elly was keeping memory to his students suddenly incorrect statement erupted and interrupted the lesson, now help him by prescribing the incorrect one among the following.

(a) Flowers of Sulpher are the fine powder of pure Sulphur

(b) Sulphur is soluble in carbon disulphide

(c) Monoclinic Sulphur is stable above 96^0C

(d) Rhombic Sulphur has needle shaped crystals [ ]

(ii) Dr. Emmanuel was demonstrating the reaction through dissolving sodium chloride in water, while the result was given out then he fail to explain the name of the process. As a chemist give out the name of the process.

A. Hydration process

B. Neutralization reaction

C. Dissolution

D. Precipitation process [ ]

(iii) What is the volume of hydrogen gas produced when 0.65g of zinc reacts completely with excess hydrochloric acid at S.T.P?

(a) 224cm^3 (b)22.4cm^3 (c)22.4dm^3 (d)2.24dm^3

(iv) Using chemistry theories and principles identify particles which tend to be produced when electrolyte are in molten state or in solution form (a) Anions (b) Cations (c) Ions(d)Molecules [ ]

(v) A form four student is doing the chemistry reaction experiment through mixing the different compounds in the test tube on doing so, the final results was obtained but he failed to identify the valences Contributed by letter T and X during writing and balancing the equation. Help him

$$TMg\ (g) + X_3N_2(s) \longrightarrow Mg_3N_2$$

(a) T is 3 while Xis 1 (b)T is 2 while X is 3 (c) T is 1 while X is 3

(d)T is 1 while X is 3 [ ]

(vi) A student teachers was in the class teaching how to write electronic configuration of different element. The electronic configuration of this element $Cl^{-1}$is ?

(a) 2.8.7 (b)2.8.8. (c)2.8.8.1 (d)2.8.6 [ ]

(vii) Mr. Osama used burning site to burn dried varieties for such habit he used to stop the spread fire using carbon gas which used to displace oxygen from burning side because?

(a) It is denser than air

(b) It does not support burning

(c) It is colorless and odorless

(d) It turns lime water milky [ ]

(viii) Mrs. Malima Spilled out concentrated acid on the floor which action should be taken immediately after concentrated acid is spilled on the skin?

(a) It should be rinsed off with large amount of running water

(b) It should be neutralized with concentrated NaOH

(c) The affected area should be wrapped tightly and shown to a medical provider

(d) It should be neutralized with solid $CaCO^3$ [ ]

(ix) Talk to Dc Sabato who left white anhydrous Copper(ii) Sulphate in air turned blue.The reasons why white anhydrous Copper (ii) Suplhate turns blue when exposed to the atmosphere is that it:

(a) Absorbs water vapor (b) React with oxygen(c) Became dry (d)Expel water to the atmosphere [ ]

(x) Two faradays (2F) were required to deposit one mole of a metallic element M. from an aqueous solution of its salts.If Element M. has no variable valiancy the empirical formula of its phosphate is:-

(a) M $(PO4)_3$ (b)MPO_4 (c)$M_2(PO_4)_3$ (d)$M_3(PO_4)_2$ [ ]

2. Match the items in LIST A with the response in LIST B by writing the letter of the corresponding response

LIST A	LIST B
(i) Calcium	A. it melts and burn with a blue flame
(ii) Magnesium	B. melts and burn with a lilac flame
(iii) Sulphur	C. burns slowly with a yellow white flame

(iv) Carbon (v) potassium	D. melts and burns with a bright white flame E. turns orange then the surface of the product turns black F. burns slowly with a dull red flame

SECTION B

Answer all questions in this section

3. (a) A form two student asked you to teach her the real meaning of fuel as applied in chemistry. How would you explain while giving two (2) gaseous fuel in industries? A doing so, state the composition of each of the gases

   (b) As a form four students what do you think on why hydrogen gas is colleted using downward displacement of water method?

   (c) (i) Madam Mwanga wanted to prepare ammonia. What are two main reagents madam Mwanga used to prepare ammonia?

   (ii) Abdallah is your best friend in your class but does not have a very quick knowledge on how concentrated Sulphuric acid becomes weaker electrolyte than dilute Sulphuric acid as an expert open the chemistry window to your friend Abdallah on his confusion.

4. A 2.12g sample of Unkwon monovalent metal carbonate (M_2CO_3)was dissolved in water to make 250cm^3 of a solution. The following results were obtained the experiment when 25cm^3 of movement metal carbonate was recorded with 0.25M hydrochloric acid solution by using methyl orange indicator.

Burette reading (cm^3)	**Pilot**	**1**	**2**	**3**
Final burette reading (cm^3)	19.00	37.90		48.10
Initial burette reading (cm^3)	3.10		16.10	
Volume of acid used (cm^3)		16.20	15.90	16.10

(a) Complete the table above and balance chemical equation for the reaction

(b) Determine the relevant atomic mass of the metal

(c) Identify the unknown element :M" in the compound M_2CO_3

5. (A) People suffering from heart burn usually use wood ashes from relief outline the characteristics which makes the ashes to be used for heart burn relief.
   (i) Give four compound found in the laboratory which show the same characteristics as ashes
   (ii) Give four applications of neutralization in our daily life
6. (a) As a form four students, briefly explain why Neon does not react with sodium
   (b) Laboratory technician allowed sodium thiosulphate solution to react with dilute hydrochloric acid, the product was deeply formed, communicate to the laboratory technician through.

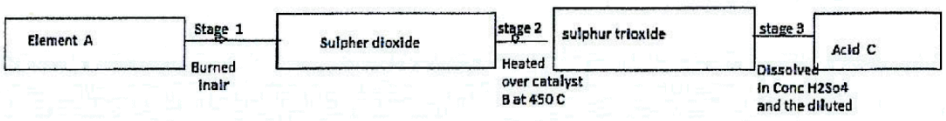
   (i) Chemistry language by a balanced chemical equation

   (ii) Providing net ionic equation for the reaction in b (i) above and state the spectator ions.

   (c) By using chemistry approaches , find oxidation number of:
   (i) Cu in $Cu(NO_3)_2$ (ii) Cl in NH_4Cl
7. (a) Vivian was sent by her father into Mangi's shop to buy of 1kg of sugar while on the way back to home the bag busted and all sugar mixed with sand
   (i) With the concept of separation of mixture explain how Vivian could obtain sugar
   (ii) Name all method of separation of mixture explain that can be involved in the above scenario
   (b) (i) By using the clear chemistry approach. Consider X and Y are isotopes where by the number of neutrons of each isotope is equal to the atomic number of the element. If the number of neutrons of X is 15 what is the mass number of X?
   (ii)If the molecular formula of calcium (ii) Sulphate is CaSO4,what is the percentage composition of oxygen in this compound?
8. (a) Briefly explain why a mixture of two liquids with equal boiling points cannot be separated by simple fractional distillation
   (b) (i) Consider the following elements having their atomic mass and atomic number 40Ar and 39K, Explain why Argon has a lower atomic number but greater atomic mass that of potassium

   (ii) When common salt react with concentrated Sulphuric acid, a gas which turn blue litmus paper red is evolved. Provide a well-balanced chemical equation for what has observed.
9. (a) Study the flow diagram showing the stages in contract process

Diagram



(i) From the stages in contact process. Give the name of element A, catalyst B and an acid C

(ii) Provide a well –balanced chemical equation for the reaction for the formation of Sulphur trioxide in stage 2

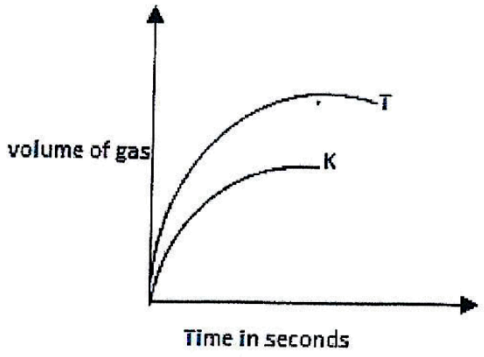
(b)Carbon dioxide can be prepare by adding an acid to calcium carbonate

(i) Using the named acid, write a balanced chemical equation for the reaction

(ii) Name all the products formed in (b) (i)

10. (a) Mr. Stephano measured the volume of gas produced when 5kg of two chemicals T and K were added separately to hydrogen peroxide under identical conditions. The results for this experiment are represented on the graph below;

Diagram



Mr. Stephano claimed that K is a better catalyst than T. His partner Suleiman did not agree

(i) Why does Suleiman think that Stephano's conclusion is wrong?

(ii) After the experiment, Suleiman recorded 5kg of K and 1g of T from the two experiments. He claimed that this shows that Mr. Stephano was wrong. Is Suleiman claim true? Give a reason.

(b)A current of 0.5A was made to flow through silver voltammeter for 30 minutes . Calculate the mass of silver deposited and the equitant weight of silver

11. (a) For each of the following reactions, identify which of the gases chlorine, Sulphur dioxide and hydrogen Sulphide is either an oxidizing agent or reducing agent. Briefly explain how you arrived to your answer.

(i) $SO_2(g)+2H2S \longrightarrow 2H_2O(1)+3S(s)$

(ii) $Cl_2(g) +2H_2O(1)+SO_2(g) \longrightarrow 2HCl(g)+H_2SO_{4(aq)}$

(c) The preparation of chlorine gas can be represented by the equation below

$MnO_2+4Hcl \longrightarrow Mn\ Cl_2+2H_2O+Cl_2$

How many moles of HCl are needed to react with 10g of MnO^2

12. Consider the following equation

$$P(g)+K(g) \rightleftharpoons T(g) \Delta H=-XkJ$$

Describe how the rate of production of T can be altered through the use of Le-Chatelier principle.